

Dear Dr. Way,

Please consider this manuscript "Phenological sensitivity to climate change disrupts species coexistence" as a Research Report in *Global Change Biology*.

1) *What scientific question is addressed in this manuscript?*

Shifts in phenology—the seasonal timing of organisms' life cycle events—are one of the most widespread biological indicators of climate change, yet have often been overlooked in ecological theory and as a result, research to date has struggled to predict how species temporal interactions will shift with climate change. Here we present a simple, yet robust modeling framework to capture the dynamic interactions between the physical environment, phenology and coexistence.

2) *What is/are the key finding(s) that answer this question?*

We integrated empirical data on seed germination phenology with a well established community assembly models to show how modeling phenological differences as the realized product of an interaction between the environment (cues) and species physiologies links environmental change to competition. With this subtle, yet essential, adjustment we demonstrate how phenologically-mediated patterns of coexistence under historical climate conditions may degrade as climate warms while providing a biologically realistic mechanism to project changes in these species iterations into the future.

3) *Why is this work important and timely?*

Our manuscript details a path forward to translate ecological theory into practice and advance a more predictive ecology than is better equipped to inform ecosystem conservation and management.

4) *Describe how your paper fits within the scope of GCB.*

Our manuscript directly addresses the effects of global climate change on plant phenology and species interactions, a topic that is relevant in natural and managed ecosystems around the world.

5) *What biological AND global change aspects does it address?*

Our paper address multiple aspects of biology including phenology, reproduction, resource acquisition and ecological dynamics in relation to global climate change.

6) *What are the three most recently published papers that are relevant to this manuscript?*

Three recent and relevant publications: Cleland & Wolkovich (2024), Diez *et al.* (2024). See next page for full citation.

7) *Provide the name and institutional email addresses of five potential reviewers.*

1. Dr. Meredith Zettlemoyer, meredith.zettlemoyer@umontana.edu
2. Dr. Mason Heberling, heberlingm@carnegiemnh.org
3. Dr. Guillaume Charrier, guillaume.charrier@inrae.fr
4. Dr. Jason Fridley, fridley@clemson.edu
5. Dr. Emily Grman, egrman@emich.edu

Sincerely,

Daniel Buonaiuto

References

- Alexander JM, Levine JM (2019) Earlier phenology of a nonnative plant increases impacts on native competitors. *Proceedings of the National Academy of Sciences*, **116**, 6199–6204. doi:10.1073/pnas.1820569116. URL <https://doi.org/10.1073/pnas.1820569116>.
- An S, Chen X, Shen M, Zhang X, Lang W, Liu G (2022) Increasing interspecific difference of alpine herb phenology on the eastern qinghai-tibet plateau. *Frontiers in Plant Science*, **Volume 13** - **2022**. doi:10.3389/fpls.2022.844971. URL <https://www.frontiersin.org/journals/plant-science/articles/10.3389/fpls.2022.844971>.
- Baskin C, Baskin J (2014) *Seeds: Ecology, Biogeography, and Evolution of Dormancy and Germination*. Elsevier Inc.
- Blackford C, Germain RM, Gilbert B (2020) Species differences in phenology shape coexistence. *The American Naturalist*, **195**, E168–E180. doi:10.1086/708719. URL <https://doi.org/10.1086/708719>. PMID: 32469665.
- Bonamour S, Chevin LM, Charmantier A, Teplitsky C (2019) Phenotypic plasticity in response to climate change: the importance of cue variation. *Philosophical Transactions of the Royal Society B: Biological Sciences*, **374**, 20180178. doi:10.1098/rstb.2018.0178. URL <https://doi.org/10.1098/rstb.2018.0178>.
- Buonaiuto DM, Wolkovich EM (2021) Differences between flower and leaf phenological responses to environmental variation drive shifts in spring phenological sequences of temperate woody plants. *Journal of Ecology*, **109**, 2922–2933. doi:https://doi.org/10.1111/1365-2745.13708. URL <https://doi.org/10.1111/1365-2745.13708>.
- Buonaiuto DM, Wolkovich EM (2023a) Contrasting responses to climate variability generate seasonal priority effects between native and invasive forest herbs. *Journal of Ecology*, **111**, 1711–1721. doi: <https://doi.org/10.1111/1365-2745.14138>. URL <https://doi.org/10.1111/1365-2745.14138>.
- Buonaiuto DM, Wolkovich EM (2023b) Germination phenology and seedling competition between two forest herb species of the eastern united states in controlled environments.
- Canadell J, Monteiro P, Costa M, *et al.* (2021) *Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*. Cambridge University Press, New York, NY.
- Chesson P (2017) Aedt: A new concept for ecological dynamics in the ever-changing world. *Plos Biology*, **15**.
- Chaine I, Régnière J (2017) Process-based models of phenology for plants and animals. *Annual Review of Ecology, Evolution, and Systematics*, **48**, 159–182. doi:https://doi.org/10.1146/annurev-ecolsys-110316-022706. URL <https://www.annualreviews.org/content/journals/10.1146/annurev-ecolsys-110316-022706>.
- Cleland EE, Allen JM, Crimmins TM, *et al.* (2012) Phenological tracking enables positive species responses to climate change. *Ecology*, **93**, 1765–1771.
- Cleland EE, Wolkovich E (2024) Effects of phenology on plant community assembly and structure. *Annual Review of Ecology, Evolution, and Systematics*, **55**, 471–492.
- Dawson-Glass E, Schiafo R, Kuebbing SE, Stuble KL (2025) Warming-induced changes in seasonal priority effects drive shifts in community composition. *Ecology*, **106**, e4504. doi:https://doi.org/10.1002/ecy.4504. URL <https://doi.org/10.1002/ecy.4504>.
- Diez J, Schlauch J, DuCharme E, Collings J, Erskine S (2024) Germination phenology alters species coexistence outcomes. *Journal of Ecology*, **112**. URL